

Gaseous ring around young star

The discovery of a constrained ring of carbon monoxide around a new star has astronomers excited.
<http://bit.ly/zzZ4WQ>

Ankle - best place to scratch

A recent study has found that scratching at one's ankle is the most pleasurable of all places to scratch.
<http://tgr.ph/wAecVb>

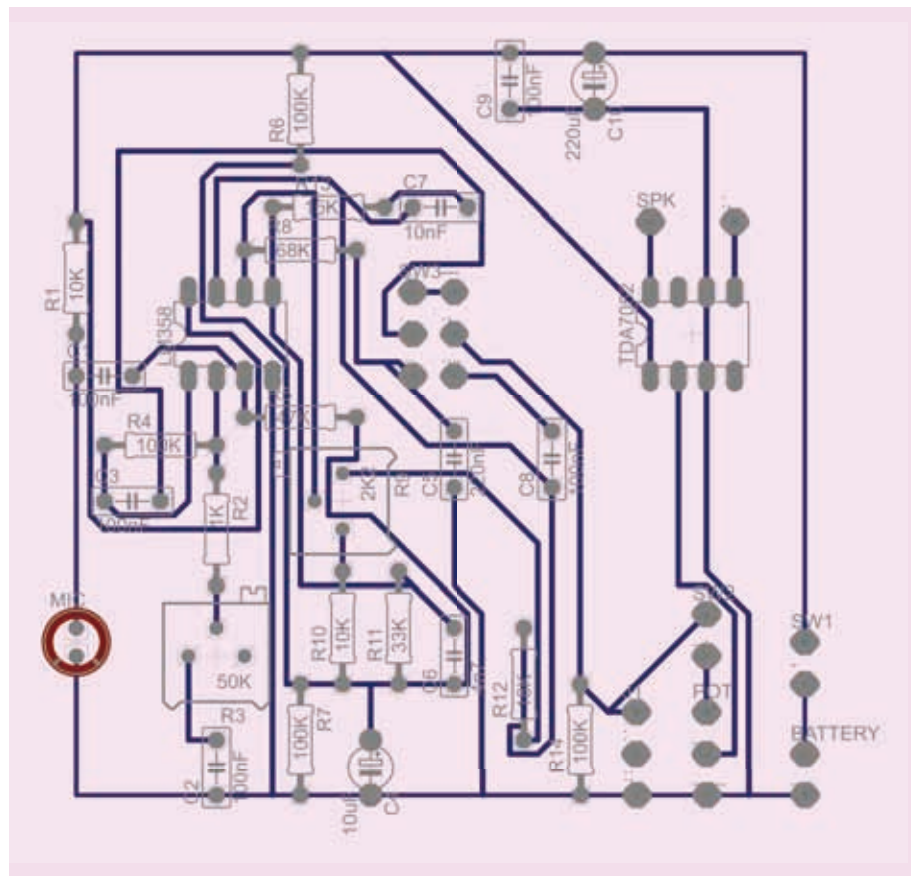
Operation

Plug in a microphone and a speaker. Switch on SW1, SW2 and SW3 and start speaking into the microphone. Vary the trimmers R3 and R9 to get the desired modulation of the voice and then vary the potentiometer P1 to increase the volume of the output. Here's what the switches do :

- SW1 - Bypasses the modulation of the sound.
- SW2 - Switches on the speaker.
- SW3 - Switches the power on/off.

Alternatives to the LM358 may be used, however the values of R11, R13, C6 and C7 need to be altered accordingly as they all together form the wein bridge. Currently the wein bridge is centred at 1 KHz. Certain alternatives like the MIC863, LT1112 and TC1029 can be used without changing the passive components. They're all pretty much the same.

If you prefer an awesome robotic voice, kind of like *James Earl Jones* doing Darth Vader from *Star Wars*, then check out ICs HT8950/WIN8072/RTS0072/PT2361. They all have four parameters that can be varied to provide a wide variety of audio signal output. With some tinkering you can generate the sounds of all



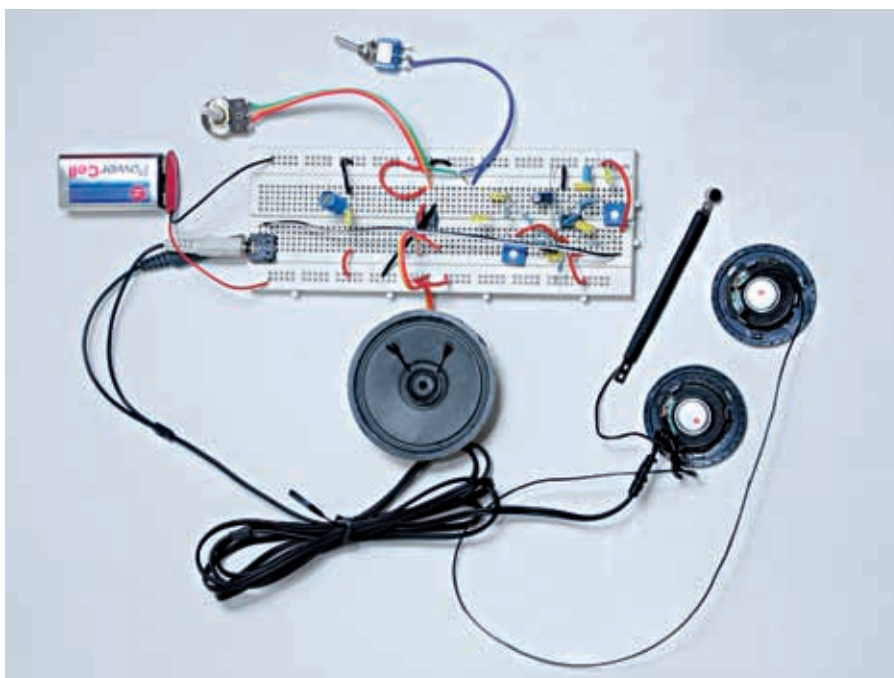
CAD diagram of the circuit

your favourite mechanical characters. However, they are not available in

India but in case you do find any, let us know. The same can be purchased on www.ebay.com but very few sellers would be willing to ship to India. Even if they did, the IC would cost about ₹200 each for low quantities. Some of these are no longer in production.

This is only a general circuit to get the job done, we hope that you'll tinker around with the design further and come up with something wacky. Do let us know about your crazy modifications. ■

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Breadboard prototype